



An Average Power Primer: Clarifying Misconceptions about Average Power and Replicability

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The replication crisis heightened interest in methods for assessing the credibility of published research. One approach is to estimate the average power of original studies based on observed data. Critics have challenged this approach as an “ontological error”, as a poor predictor of replication outcomes, and as too imprecise to be useful. This article aims to address these critics. We respond by clarifying that using observed data to estimate true power is a standard inferential practice and does not constitute an ontological error. The goal of average power estimation is not to predict the outcome of future replication studies, but the hypothetical outcome if original researchers had to replicate their studies with new samples. Lastly, we demonstrate that even with substantial uncertainty, average power estimates remain diagnostically informative, especially under selection for statistical significance. Using a z-curve analysis of terror management research, we illustrate that a corpus of significant findings does not, by itself, imply evidential value under severe selection for significance. We conclude that, despite limitations, average power estimation remains a valid tool for evaluating published research.

Keywords: statistical power, replicability, meta-analysis, credibility, uncertainty

During the 2010s, the field of psychology became increasingly aware of the difficulty in replicating many published findings. The most notable demonstration of replication problems is the Reproducibility Project (Open Science Collaboration, 2015). In this project, 100 studies published in three journals were replicated as closely as possible and only 36% of the replication studies obtained a statistically significant result. The success rates were slightly higher in cognitive psychology (50%) than in social psychology (25%) (Open Science Collaboration, 2015). Efforts have been directed towards assessing the replicability of published results in psychology. In some cases, even large-scale replication studies with thousands of participants failed to replicate results despite hundreds of significant results in peer-reviewed journals (Vohs et al., 2021). The underwhelming results of seemingly robust findings supported by dozens of significant results have led some researchers to declare a replication crisis or a crisis of confidence in psychological science (Schimmack, 2021; Vazire, 2018).

In response to the replication crisis, psychologists began to examine scientific practices that may increase the likelihood of publishing a false-positive result (John et al., 2012). In parallel, several statistical tools were developed to diagnose publication bias, questionable research practices, and evaluate evidential value while

accounting for selective reporting, such as the Test of Excess Significance (TES) (Francis, 2014; Ioannidis and Trikalinos, 2007; Schimmack, 2012), p-curve (Simonsohn et al., 2014), and z-curve (Bartoš and Schimmack, 2022; Brunner and Schimmack, 2020). Although these approaches differ in their assumptions and estimands, these methods share a common goal: using the published record to assess whether a literature is consistent with unbiased reporting and plausible underlying (true) power.

A few critical articles have questioned the validity and usefulness of retrospectively estimating underlying average power from published findings (e.g., McShane et al., 2020; Pek et al., 2024). These critiques raise three major concerns: an alleged ontological error when applying statistical power concepts post-study for evaluation, limited predictive validity of average power for future replication outcomes, and substantial uncertainty in average power estimates. In this article, we carefully address these criticisms. First, we clarify the inferential target of retrospective use of statistical power to evaluate completed studies and explain why limited single-study predictive accuracy does not undermine its value for diagnosing evidential value in published literature. That is, evidence of publication bias and low evidential value in published studies can reveal the need for high-powered replication studies. It is therefore important

to address concerns that evaluations of published studies are theoretically or practically impossible. Average power allows us to use statistical power as a descriptive diagnostic (post-study) rather than solely as a design parameter (pre-study). Second, we argue that average power estimates, together with uncertainty intervals, remain informative for evaluating credibility. Third, we illustrate with a case study how average power can assess the evidential value even in a literature with a high discovery rate, and how the resulting estimates can inform replication planning.

Ontological Clarification: Pre-Outcome and Post-Outcome Probabilities

Traditionally, statistical power is defined as the long-run probability that a study yields a statistically significant result for a given population effect size (Cohen, 1988). Conditioning on a non-zero effect size makes sense for a priori power analysis, but it is less suitable for estimating the average probability of significance across heterogeneous literatures, because some studies may have true effects of zero. It is therefore important to distinguish between hypothetical power and true power. True power is the power corresponding to the true population effect size, but it is unknown because population effect sizes can only be estimated with sampling error. Nevertheless, like other population parameters, true power can be estimated from data by using an effect-size estimate as input to the power function (Yuan and Maxwell, 2005). These estimates are typically called observed power. Although “observed” can be misread as implying the true value, we retain the term and explicitly incorporate sampling error when interpreting these estimates.

We distinguish three types of power: hypothetical power, $\epsilon_H = f(\alpha, N, \theta_H)$, observed power, $\hat{\epsilon} = f(\alpha, \hat{N}, \hat{\theta})$, and true power, $\epsilon_T = f(\alpha, N, \theta_T)$. The subscript “H” denotes a hypothetical population parameter that is neither true nor observed such as ones used for study planning, whereas “T” denotes the true (but unknown) population parameter, (H = Hypothetical, T = True).

Hypothetical power (ϵ_H) describes a power estimate where θ_H is a speculation of the population effect size. Hypothetical power is often used to plan sample sizes during the pre-study phase using an assumed true population effect size. Observed power ($\hat{\epsilon}$) is estimated plugging the observed effect size (and, where relevant, its uncertainty) into the power function. The true power (ϵ_T) represents the long-run probability of obtaining a significant result, based on the unknown population effect size (θ_T). Table 1 summarizes each of the power calculations shown above.

We refrain from using terms such as a priori and post-

hoc power because power calculations can be conducted before and after a study. Moreover, those conducted after a study can use hypothetical values or observed data. We also do not discuss the problematic use of observed power to evaluate the results of a single study (Hoenig and Heisey, 2001). Our focus is on using observed average power from a set of studies as estimates of the true average power of these studies.

Pek et al. (2024) argue that it is a fatal ontological error to estimate the power of a study after the study has been completed, “Applying a probability over random data to fixed (observed) data is a fatal ontological error” (p. 5). We agree that an observed outcome (e.g., a significant p-value) is a fixed fact. However, inferential statistics routinely use observed data to estimate unknown population parameters. This fixed result was generated by a stochastic process with unknown success probability, the study’s true power. The true power therefore remains unknown and can be inferred.

Inferential Use of Statistical Power

Average power calculations do not make claims about the probability of already completed study outcomes. For example, the original studies in the Reproducibility Project had 97 significant results, which is typical for psychology journals (Sterling et al., 1995; Sterling, 1959). Estimating the true power of these studies is not a prediction of the probability of the completed studies to produce significant results because the percentage of significant results is known to be 97%. The goal is not to assign a probability to the set of realized studies with a fixed outcome, but to estimate the expected success rate if the same study were repeated under identical conditions with a new random sample. This is conceptually analogous to estimating the true proportion of any binary outcome (heads vs. tails of a coin toss) on the basis of a sample of events (Brunner and Schimmack, 2020).

A fair coin has a 50% probability to obtain heads or tails. After tossing the coin, the outcome is either heads or tails. The true power parameter of the coin determines the long-run probability of that coin to produce heads again in the next toss or the long-run frequency of heads and tails of the coin. We cannot infer this coin’s long-run probability of drawing heads (ϵ) from a single coin toss ($\hat{\epsilon}$), in the same manner that we cannot estimate the true population effect size (θ) present in a study from the single observed effect size ($\hat{\theta}$). If we continue randomly selecting and tossing coins from different populations with different probabilities, we will have a series of observed outcomes. Then, the observed rate of heads $\hat{P}(\text{Heads})$ can be used to estimate the mean probability to draw heads if we were to toss all the coins

Table 1

Types of Statistical Powers Based on Input Parameters

Type of Power	Formula	Description	Timing
Hypothetical Power	$\epsilon_H = f(\alpha, N, \theta_H)$	Power calculated using hypothetical (assumed) parameters that are neither true nor observed.	Pre-Study
Observed Power	$\hat{\epsilon} = f(\alpha, \hat{N}, \hat{\theta})$	Power calculated using observed statistics from actual data as parameters.	Post-Study
True Power	$\epsilon_T = f(\alpha, N, \theta_T)$	Power calculated using the true population parameters.	Post-Study

Note. The subscript “H” refers to a parameter hypothetical in nature, and subscript “T” refers to a true parameter.

again, the population mean true power ($\bar{\epsilon}_T$). Therefore, it is also possible to estimate the probability of drawing heads if we were to toss another coin from the exact same populations where we drew for the original coin set. The same logic can be applied to published research, the observed success rate of a set of studies $\hat{P}(\text{Significant})$ can be used to estimate the population mean true power ($\bar{\epsilon}_T$), that is, the long-run probability of the studies to produce the observed outcome. A single study, like a single coin toss, does not provide enough information to estimate the true probability of a significant outcome or to evaluate whether a coin is fair. In contrast, a sample of outcomes makes it possible to estimate underlying probabilities—even when the events have different probabilities (i.e., studies with different true powers or coins with different biases) (Brunner and Schimmack, 2020). Thus, estimation of average power does not make the implausible assumption that all studies have the same power.

Power-based tests of publication bias rely on the fact that publication bias inflates the rate of significant results more than it inflates average power (Schimmack, 2012). This makes it possible to demonstrate empirically that publication bias and selective reporting contribute to the high success rates in psychology (Sterling et al., 1995; Sterling, 1959). In the Reproducibility Project, the original studies yielded 97 significant results, but actual replication studies produced only 36% significant results (Open Science Collaboration, 2015). This finding suggests that the true average power of the original studies was only about 40%, and the 97% success rate was obtained with questionable research practices (John et al., 2012). Power calculations make it possible to demonstrate these biases without actual replication studies.

In summary, it is not an ontological error to infer power from the results of observed studies. It would be an ontological error to use this information to make

claims about the completed studies, but it is not a mistake to use this information to estimate the outcome of exact replications with new samples.

Average Power and Replicability

Another concern has been “average power cannot quantify the replicability of an effect” (Pek et al., 2024, p. 11). It is often assumed that the purpose of power estimation is to predict outcomes of real-world replication studies, which are often non-exact due to changes in context, samples, or procedures (McShane et al., 2020). All these deviations from the original study influence the outcome of an actual exact replication study. However, this argument does not hold for hypothetical exact replication studies that by definition are identical to the completed studies, including contextual variables (LeBel et al., 2018). We agree that it is not very valuable to predict the outcome of actual exact replication studies, especially when average power of the original studies is low. In this case, it would be futile to replicate the original studies exactly with the same sample sizes. Rather, replication studies would want to increase sample sizes and might also improve methodological weaknesses of original studies (e.g., Doyen et al., 2012).

However, predicting outcomes of new replication studies is not the primary goal of estimating average power. The key question is what results one would expect in a hypothetical exact replication project where the original authors redo their studies exactly as they were done, but with a new sample. This information is particularly interesting when results are selected for significance. When selection for significance is present, statistical significance provides no information about the false positive risk (Sterling, 1959). Estimates of average power, however, make it possible to distinguish credible results that were driven by true effect sizes from sets of studies with low average power that may contain a large percentage of false positive re-

sults (Sorić, 1989). In short, average power estimation provides a diagnostic tool for evaluating the evidential value of sets of studies.

Real replication studies can, however, be used to estimate replicability, if they are reasonably similar to the original studies. The original studies used for the Reproducibility Project had a 97% success rate (Open Science Collaboration, 2015). Taken at face value, this finding implies that the studies had an average power of 97%. However, the replication studies had only a success rate of 36%. This suggests that the replication studies had an average power of 36%. The main assumption here is that differences between original and replication studies are unrelated to the true power of the original studies. Statistical analyses of the original studies do not require this assumption because their estimates are based on results of the original studies. Convergent evidence across these different approaches provides strong evidence that high success rates are not justified by high power and instead suggest the use of questionable research practices.

The issue with the original estimate is that it does not take into consideration that just as the average observed effect size can be biased by selective publication, the observed success rate can be biased as well. It is imperative that selection bias is accounted for when estimating average power, as selection bias systematically inflates the observed success rate. Thus, we cannot use the observed success rate of a set of studies at face value.

Addressing Uncertainty in Average Power Estimates

Pek et al. (2024) draw on McShane et al.'s (2020) simulation studies to argue that estimates of average power are too variable and imprecise to be useful. They cite McShane et al.'s (2020) article to support this claim, but McShane et al. (2020) points out that no fixed amount of uncertainty invalidates empirical results, "Although what constitutes a tolerable degree of variability varies by context, we view a 95% sampling distribution width of .2 as the worst tolerable for estimating average power" (p. 191).

We agree that uncertainty is a concern, especially in small samples or highly heterogeneous literatures. However, uncertainty does not negate the informational value of power estimates. For example, if an average power estimate ranges from 10% to 40%, we can safely conclude that the original studies are underpowered and that future studies should increase sample sizes to avoid false negative results, even if there is uncertainty about the true power of the studies. From this finding, we can also infer that statistically significant effects observed in such settings are likely to be inflated, because low true power requires larger observed effects

to achieve significance. This conclusion follows from the statistical fact that a two-sided p-value equals alpha (.05) corresponds to 50% power for a symmetric test. Thus, a study with 40% power requires an inflated effect size estimate to achieve significance.

The main problem with estimates of observed power is the fallacy of confusing observed and true parameters, and the term "observed" does not help. Point estimates of average power are not observations of the true average power. To avoid this confusion, all estimates of true average power should be accompanied by uncertainty indicators such as 95% confidence intervals. The estimation error shrinks as the number of studies increases, just as sampling error shrinks with the number of participants in a study. Uncertainty can sometimes be less than McShane et al.'s (2020) arbitrary criterion of 20 percentage points and sometimes more. Whether the results are useful depends on the research question and the actual width of a confidence interval. For example, an EDR of 10%, 95% CI [5%-17%] and an EDR of 26%, 95% CI [14%-55%], both tell us most studies are underpowered, despite the width of the confidence intervals. If we come across a set of studies with an EDR of 35%, 95% CI [5%-85%], we cannot safely conclude that all studies were sufficiently powered and future research should increase sample sizes to avoid false negatives. More generally, the usefulness of true power estimates depends on the question being asked and the precision of the available estimates.

In conclusion, uncertainty in estimates of average power can be quantified with confidence intervals and average power estimates can be useful even when uncertainty is substantial. The following section will use terror management research to showcase how average power can be useful to evaluate the credibility of completed studies.

Case Study Application: Terror Management Meta-Analysis

We focus on z-curve for this re-analysis because the test of excess significance (TES) only evaluates whether the number of significant findings exceeds expectations and does not provide a direct estimate of average true power. We also do not use p-curve, which assumes that all studies share a common power level; this assumption is routinely violated in meta-analyses of heterogeneous studies and can lead to biased estimates (Brunner and Schimmack, 2020). In contrast, z-curve was designed with heterogeneity in mind: it explicitly models variation in underlying power and makes no assumptions about the shape of the power distribution across studies.

Chen et al. (2025) published a meta-analysis of over

800 terror management studies. Terror management studies use primes of death to examine the effect of mortality salience on people's beliefs, attitudes, and behaviors. Their article included a z-curve analysis but cautioned that the results might be uninformative, echoing Pek et al.'s (2024) concerns about estimating power from completed studies. We are concerned that such statements may encourage readers to dismiss z-curve results outright.

We reproduce the z-curve analysis reported in their article here and explain for readers unfamiliar with z-curve how results of a z-curve analysis can be used to evaluate the credibility of published results. The data used for this case study were made publicly available by Chen et al. (2025) and the code to replicate the z-curve results is available in an associated OSF repository, <https://doi.org/10.17605/OSF.IO/7CK4V>. The re-analysis of the data was not preregistered, but it does not deviate from the analyses reported by Chen et al. (2025). The only difference is that we use z-curve 3.0 (Schimmack, 2025), rather than the zcurve R package (Bartoš and Schimmack, 2026), because z-curve 3.0 contains developmental features not yet implemented in the package. The package provides information about heterogeneity in power by estimating local power for subsets of studies with different z-scores; these estimates appear below the x-axis in the plot.

Z-curve is a selection model that uses the distribution of z-scores to estimate average power. Unlike the meta-analytical approaches criticized by Pek et al. (2024) that estimate average power from a meta-analytic effect size, z-curve estimates the population mean true power of a random sample of tests that have been selected for significance. By modeling the distribution of underlying the true power values across studies, the model accounts for heterogeneous true effect sizes without assuming the true distribution and selection bias (see Bartoš and Schimmack, 2022; Brunner and Schimmack, 2020, for further details).

On visual inspection, a peak at $z = 1.96$, a steep decrease, and few studies with high z-scores ($z > 4$) imply that average power is low. Studies with 80% power would have a peak at 2.8. Z-curve provides an estimate of average power by fitting a model to this distribution and assuming a simple selection process, non-significant results are not reported, whereas significant results are. Chen et al. (2025)'s dataset only contained significant results, so no observations appear to the left of the significance criterion. However, this pattern is not unique to their coding decisions: psychology journals have long shown extremely high success rates, typically above 90% (Fanelli, 2012; Sterling, 1959), with more recent estimates still exceeding 70% (Schneck, 2023).

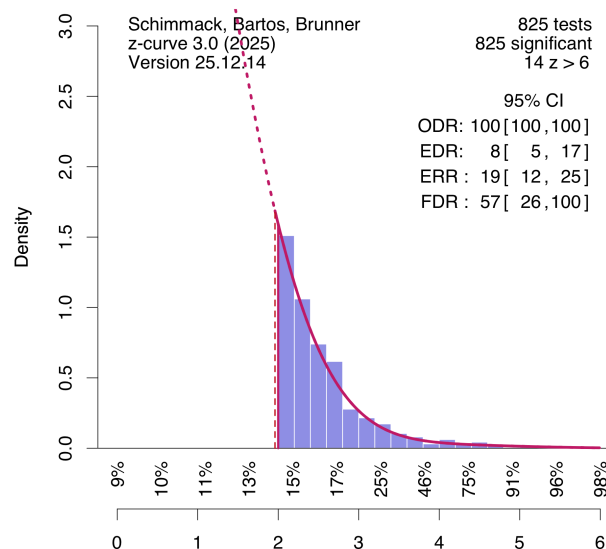


Figure 1

A Z-curve plot for Terror Management Studies

Thus, the figure may exaggerate selection bias, but the distribution of significant z-values makes it unlikely that all the implied non-significant results were reported. Z-curve produces several quantitative estimates to evaluate the credibility of a literature (Table 2).

Z-curve accounts for heterogeneous effect sizes and sample sizes. The model assumes that significant results are likelier to be published, therefore, it remains assumption-bound, and it may not recover the true underlying distribution with certainty. The method assumes that the only distortion is selection for statistical significance. It assumes that p-hacking, data-dependent stopping, or other practices that would change the distribution of significant results were not used or do not substantially alter the distribution of significant results. All significant results are assumed to be reported without further bias. With this assumption, it is possible to estimate the average power of all studies that were conducted, including the non-significant results that are missing in Figure 1. The Expected Discovery Rate (EDR) is an estimate of the average true power of the studies that were conducted, including studies that produced non-significant results that may not be reported. The point estimate is 8%, as reported in Chen et al. (2025). Inspection of the confidence interval shows that the margin of error is less than 10 percentage points. The confidence interval is also asymmetrical because power cannot be lower than $\alpha = 5\%$. The primary conclusion from this finding is that the true power of most studies in this literature is extremely low,

Table 2

Z-curve Estimates

Estimate Name	Description
Expected Discovery Rate (EDR)	Estimate of the actual percentage of significant results without selection for significance. The population mean true power before selection for significance.
Observed Discovery Rate (ODR)	Estimate of the observed rate of significant results published.
Expected Replication Rate (ERR)	Estimate of the average power of published significant results. The population mean true power after selection for significance.
False Discovery Risk (FDR)	Estimate of the maximum false positive risk corresponding to the estimated EDR.
Selection Bias (ODR – EDR)	Dataset contains more significant results than expected based EDR.

Note. Each name is accompanied by their corresponding acronym.

and that we would expect many non-significant results. However, the observed discovery rate (ODR) is 100%. Even if some selection bias was introduced by the meta-analysts, we know that success rates in this field are over 90% (Schimmack, 2021). Thus, we have strong evidence of extreme selection bias ($ODR - EDR > 70$ percentage points).

The EDR can also be used with the formula developed by Sorić (1989) to estimate the maximum false discovery rate that is compatible with the estimated EDR. Schimmack and Bartoš (2023) refer to this estimate as the False Discovery Risk (FDR). The relationship between EDR and FDR is not linear. The lower limit of the EDR confidence interval is 5%, which implies that the maximum FDR is 100% (i.e., all significant results could be false positives). At $\alpha = .05$, chance alone produces about 1 significant result out of 20 tests. Sorić's (1989) formula makes it possible to quantify the maximum FDR for other EDR values. The upper limit of the confidence interval is 17%. This still implies that the maximum FDR is about 100%. The range of possible FDR values is wide, ranging from 26% to 100%, with a point estimate of 57%. The important empirical conclusion is that more than 800 studies are consistent with the possibility that the significant evidence could be entirely due to chance. Moreover, confidence intervals around the FDR are not always this wide. Therefore, high uncertainty about the true FDR is itself informative. A credible set of studies should have false positive risks below 5% (Schimmack and Bartoš, 2023).

Alternatively, it is possible to focus on the average power of the published studies with significant results. When studies are heterogeneous, the significance filter favors studies that have higher power because they are more likely to produce significant results. Thus, the av-

erage power of studies with significant results can be higher than the EDR (Brunner and Schimmack, 2020). The average true power of studies with significant results is called the Expected Replicability Rate (ERR) because it estimates the outcome of exact replications of the original studies with significant results if they were repeated with new samples, holding everything else constant.

The ERR is 19% with a confidence interval of 12% to 25%. This confidence interval shows that average power estimates can be precise in large meta-analyses and contradicts claims that they are always too wide to be useful.

The confidence interval of the ERR does not include a value of 5%. Thus, the ERR provides a less pessimistic perspective than the conclusion based on the EDR that all studies could be false positives, because it indicates that the selected significant studies are expected to replicate more often than chance. However, the upper limit of the ERR also shows that average power is low and that replication studies with the same sample sizes are more likely to produce non-significant results than significant ones. This is valuable information because it means that future studies need to use much larger samples to provide meaningful tests of the theory.

An ERR higher than the EDR indicates that studies vary in power. That is, if all studies had an average power of 10%, selection for significance could not select for more powerful studies. Heterogeneity in power also implies that some studies are more likely to produce a significant result in an actual replication study than others. To capitalize on this heterogeneity, it is possible to lower the criterion for significance to reduce the false positive risk (Soto and Schimmack, 2025a).

Z-curve provides information about the local power of studies. Notably, studies with z-scores between 2.0 and 2.5 have only about 15% power. Acceptable levels of average power are not reached until z-scores exceed 4. At the upper end of the distribution, there are 15 studies with z-scores greater than 6, which correspond to nearly 100% power. Terror-management researchers may wish to examine these high-powered studies more closely to identify potential moderators that distinguish them from the rest of the literature. Without a review that uses statistical power as a criterion for credibility, one could mistakenly assume that a standard effect-size meta-analysis that does not take publication bias into account provides credible evidence about this literature.

Conclusion

In this commentary, we addressed three criticisms of statistical methods that estimate average power to assess the credibility of published results. First, we showed that the ontological error argument is invalid because it conflates observed outcomes with the underlying probability of these outcomes. The observed outcomes of completed studies are fixed facts, but power is a property of a data-generating process. The observed power of a set of completed studies is not a probability. Maybe the term “observed” is misleading. We do not observe the true average power. What we observe are biased estimates of population effect sizes and statistical power. When these observed data are used to estimate true power, true power remains a probability because we did not observe the true power. We merely have an estimate of true power based on observed data. This distinction between estimates and population parameters is often blurred, but crucial. The observed outcomes of completed studies are fixed facts, but power is a property of a data-generating process. Once we make a clear distinction, the ontological error argument vanishes. A coin flip can produce heads and the probability of this outcome can be 50%; both statements can be true, and observing the outcome in a single coin flip does not alter the probability of this outcome.

Second, we clarified what average power can and cannot speak to about replicability. Average power is directly tied to the expected success rate of hypothetical exact replications that repeat the original procedures and sample sizes and differ only in sampling error. It is not intended to predict the outcome of inexact replications that change populations, procedures or sample sizes. Accordingly, arguments that real-world replications are rarely exact are irrelevant to the use of average power as a diagnostic for evaluating the credibility of completed studies. This conclusion directly contradicts Pek et al.’s (2024) claim that power should be used

only for planning future studies and never for evaluating completed studies. We argue that evaluating completed studies is precisely where average power is most informative, especially when publication bias renders statistical significance uninformative and inflates effect-size estimates. In contrast, hypothetical power calculations do not assure credible results because researchers can justify the use of small samples by assuming large effect sizes. Thus, true power is much more important for evaluating credibility than a priori power calculations, which depend on assumed effect sizes and can be disconnected from the evidential value of the completed studies.

Third, we addressed the argument that average power estimates are too imprecise to be useful. Uncertainty is unavoidable and depends on features of the evidence base (e.g., the number of studies and the degree of heterogeneity). The relevant question is not whether uncertainty exists, but whether the resulting confidence interval is informative for the decision at hand. Even McShane et al. (2020) acknowledged that tolerable variability depends on context. We further demonstrated this with an estimate of the average power of terror management studies. Despite the hundreds of significant results, the estimated Expected Discovery Rate is very low, implying that many non-significant results would be expected in the absence of selection. The width of the confidence intervals is a function of the number of studies, and the usefulness of estimates must be evaluated on a case-by-case basis. The same example shows why uncertainty can be informative rather than disabling. In terror management research, the 95% confidence interval for the false discovery risk includes values as high as 100%, which undermines claims that a large number of significant findings necessarily implies strong evidential value. More generally, point estimates should not be interpreted as population parameters. For example, an estimated false discovery risk of 58% should not be read as showing that 42% of significant results are true positives; the upper bound of the confidence interval is critical for evaluating worst-case false-positive risk. A credible body of literature should yield a low false discovery risk, ideally with an upper confidence bound in a range consistent with low false-positive concern (e.g., below 5% under conventional standards) (Schimmack and Bartoš, 2023).

Z-curve is not perfect, and it is still developing. Related approaches are also being proposed (van Zwet et al., 2023). More research on the method and validation with real data is welcome. Further theoretical work, simulation studies, and validation with real data will improve our ability to evaluate published results.

Nevertheless, z-curve has already been used to assess the credibility of published results in several fields with mixed results. Some authors have documented statistical evidence of publication bias in dishonest research (Bartoš, 2024), gambling intervention tools (McAuliffe et al., 2021), system justification theory (SJT) (Crede and Sotola, 2024) and Construal Level Theory (CLT) (Maier et al., 2022), while others have documented little to no evidence of publication bias in technology education (Buckley et al., 2023) and mental fatigue and exercise (Holgado et al., 2023). Z-curve analyses have been used to assess the credibility based on statistical power estimates of results published in journals such as *Cognition & Emotion* and *Emotion* (Soto and Schimmack, 2025b), *Journal of Sports Science* (Mesquida et al., 2023), organizational journals (Crede and Sotola, 2024), personality journals (Sotola and Credé, 2023) and, there is an forthcoming review of the 10 most impactful journals in occupational health (Veen et al., 2024). Additionally, independent comparisons between z-curve estimates to large-scale replication efforts have shown that average power can be a valid predictor of actual replication outcomes (Sotola, 2023). Actual success rates tend to be lower than the ERR, because the ERR assumes exact replications but actual replication rates are higher than the EDR, because selection for significance does select for studies with higher true power. Other potential uses for average power estimations include evaluating the efficacy of policy changes in addressing selection bias over time as seen in Soto and Schimmack's (2025a) evaluation of open-science practices.

In sum, the main objections raised against retrospective estimation of average power reflect misunderstandings about what is being estimated, what the estimand implies about replicability, and how uncertainty should be interpreted. Clarifying these points helps researchers, reviewers, and readers interpret z-curve and related methods appropriately and respond to criticisms that rest on conceptual errors.

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Author Contributions

Maria D. Soto: Conceptualization, Data curation, Formal analysis, Software, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing

Ulrich Schimmack: Conceptualization, Funding acquisition, Methodology, Supervision, Writing – Review & Editing

Open Science Practices



This article earned the Open Data and Open Code badge for making the data and code openly available. It has been verified that the analysis reproduced the results presented in the article. The entire editorial process, including the open reviews, is published in the online supplement.

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